ModelIt!

Student Activity Pack

Why Are Coral Reefs Dying?

Modeling Ocean Acidification, Thermal Bleaching, and Ecosystem Collapse

Grade 12L2 | NGSS HS-LS2-6 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you know about coral reefs? Are they plants, animals, or something else? Where in the world do you find them?**

2. **Have you heard the term coral bleaching? What do you think causes it, and why is it a problem?**

3. **How do you think the health of an underwater ecosystem like a coral reef could be connected to the carbon dioxide we emit on land?**

4. **Why might losing coral reefs matter to people who live nowhere near the ocean?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sea Surface Temperature | Ocean pH Level | Coral-Zooxanthellae Symbiosis | Coral Calcification Rate | Reef Biodiversity Index | Algae Overgrowth | Herbivore Fish Population

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate Warming

Sea Surface Temperature: +1C Above Max | Ocean pH: 8.1 (Current) | Herbivore Fish: Current

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Double Stress

Sea Surface Temperature: +1.5C Above Max | Ocean pH: 7.9 | Herbivore Fish: Current

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Managed Resilience

Sea Surface Temperature: +1C Above Max | Ocean pH: 8.1 | Herbivore Fish: High (Protected)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Nutrient Pollution Runoff — Excess nitrogen and phosphorus from agriculture and sewage that enter coastal waters — these nutrients fuel algae growth that smothers coral, blocks light, and consumes oxygen, adding a local stressor on top of global ocean warming and acidification

Coral Genetic Diversity — The range of heat-tolerance genes present in the coral population — more genetically diverse reefs have a greater chance that some individuals can survive warming events, while genetically uniform reefs face total loss from a single heat wave

Marine Protected Area Coverage — The percentage of the reef under legal protection from fishing, development, and pollution — protected areas maintain higher fish populations, lower nutrient pollution, and greater coral cover, increasing overall reef resilience to climate stressors

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Coral reefs support 25% of all marine species but cover less than 1% of the ocean floor — so what happens to a quarter of ocean life when rising temperatures and acidification push these irreplaceable ecosystems past their breaking point? Use evidence from your simulation to support your answers.

1. **How does the concept of phase shifts in reef ecosystems parallel tipping points in the climate system you learned about in the previous lesson?**

2. **Why is the frequency of bleaching events more important than the severity of any single event? What does your model show about recovery time?**

3. **How do local stressors like overfishing and pollution interact with global stressors like warming and acidification? Can addressing local stressors save reefs?**

4. **What does the loss of a coral reef mean for the 25% of marine species that depend on it? Trace the cascading effects through the food web.**

5. **What are the ethical implications of a warming world where wealthy nations that contributed most to emissions can afford reef restoration while developing nations that depend on reefs for food and income cannot?**

8 STEM Challenge: Design a Coral Reef Rescue Strategy

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YOUR ENGINEERING MISSION

Design a multi-scale intervention plan that combines local reef management with global advocacy to maximize the survival chances of a threatened coral reef ecosystem.

THE SCENARIO

A Caribbean coral reef that once supported a thriving fishing community has lost 60% of its coral cover in the past 15 years due to repeated bleaching events, overfishing of herbivorous fish, and nutrient pollution from coastal development. The remaining coral is weakened, algae coverage is increasing rapidly, and the local fishing industry is collapsing. A marine conservation organization has hired your team to design a rescue plan that addresses both immediate local threats and long-term global stressors. Your plan must be scientifically grounded in the reef dynamics your model revealed.

Engineering Design Thinking

1. WHAT SPECIFIC LOCAL MANAGEMENT ACTIONS DOES YOUR PLAN IMPLEMENT FIRST, AND HOW QUICKLY CAN THEY TAKE EFFECT?

2. HOW DOES YOUR PLAN ADDRESS THE POSITIVE FEEDBACK LOOPS THAT LOCK IN ALGAE DOMINANCE AND PREVENT CORAL RECOVERY?

3. WHAT ROLE DOES CORAL RESTORATION (TRANSPLANTING, SELECTIVE BREEDING FOR HEAT TOLERANCE) PLAY IN YOUR STRATEGY?

4. HOW DOES YOUR PLAN ACCOUNT FOR THE CONTINUED INCREASE IN OCEAN TEMPERATURE AND ACIDIFICATION OVER THE NEXT 30 YEARS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Marine biologists who specialize in coral reef ecology study the complex interactions that maintain reef health and develop conservation strategies. They work for NOAA, marine sanctuaries, universities, and conservation organizations, earning $55,000-$110,000/year. Coral restoration specialists who develop and implement reef rehabilitation techniques earn $50,000-$95,000/year. Marine policy analysts who translate reef science into fishing regulations and marine protected area designations earn $60,000-$120,000/year.